

Polyalphabetic Substitution Cipher

4. Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution

cipher for each successive letter of plaintext, depending on a key.

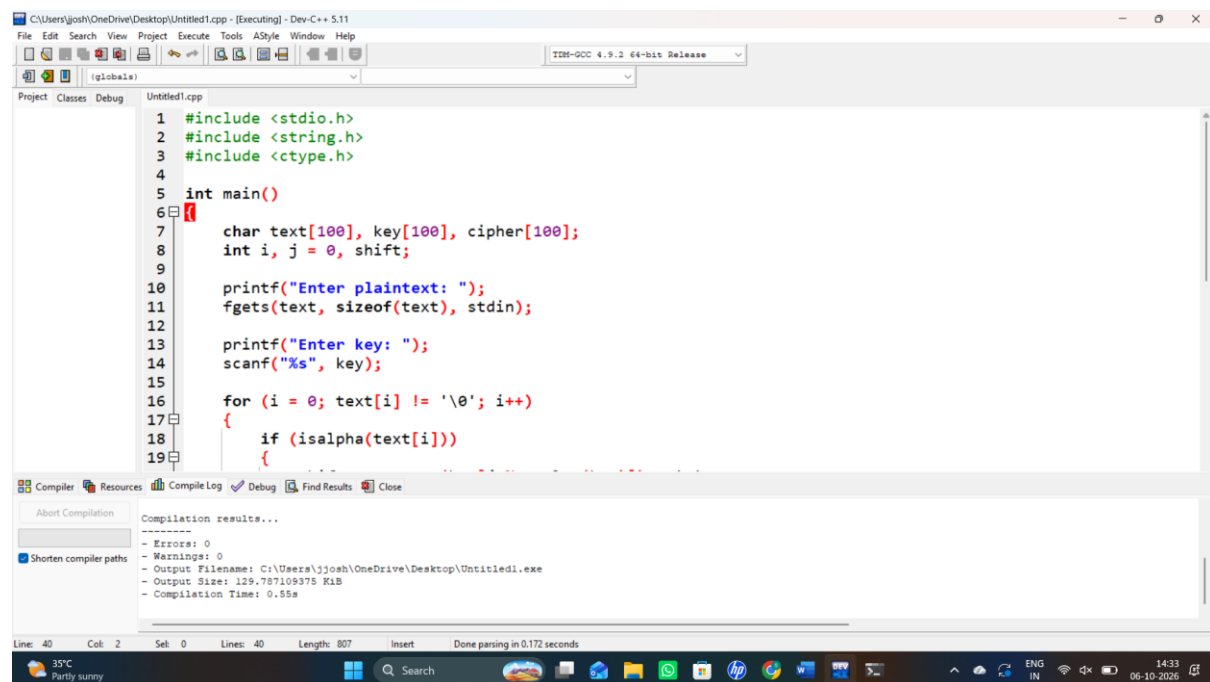
Aim

To write a C program to implement a **polyalphabetic substitution cipher**, where different Caesar shifts are used for successive plaintext letters based on a key.

Algorithm

1. Start the program.
2. Read the plaintext and key.
3. Convert each letter of the plaintext to a number from 0 to 25.
4. Convert each letter of the key to a number from 0 to 25.
5. For each plaintext letter, add the corresponding key value.
6. Apply modulo 26.
7. Convert the resulting value back to a letter.
8. Display the ciphertext.
9. Stop.

Program



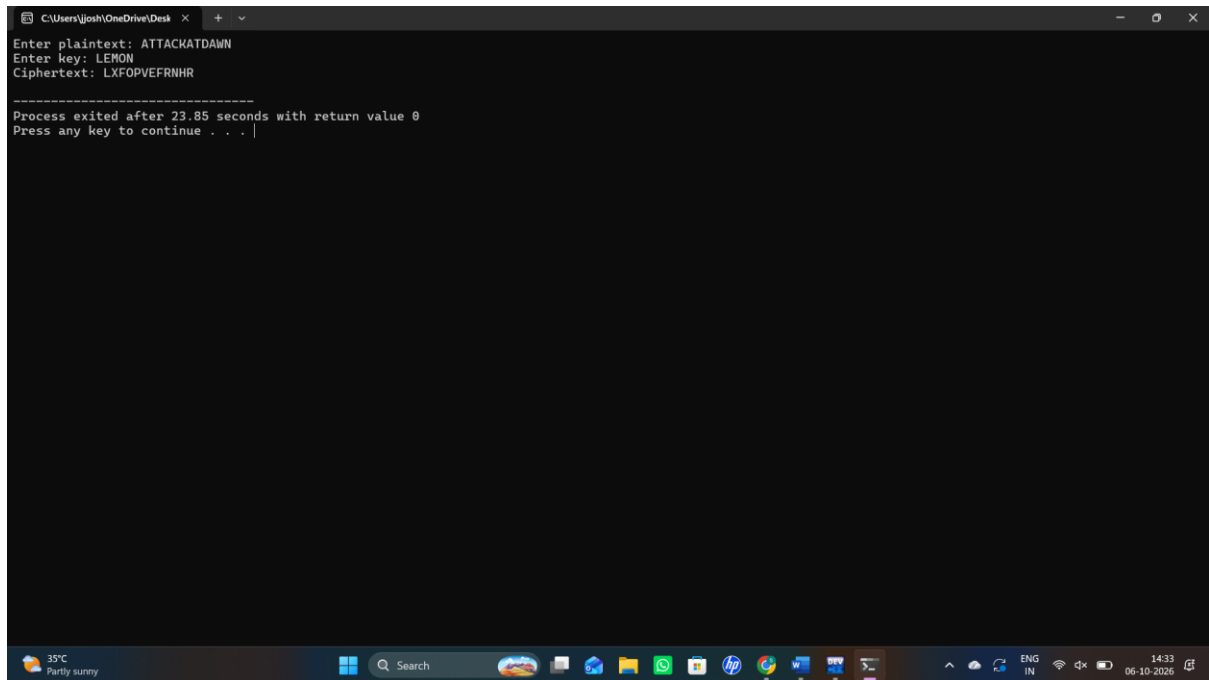
```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main()
6 {
7     char text[100], key[100], cipher[100];
8     int i, j = 0, shift;
9
10    printf("Enter plaintext: ");
11    fgets(text, sizeof(text), stdin);
12
13    printf("Enter key: ");
14    scanf("%s", key);
15
16    for (i = 0; text[i] != '\0'; i++)
17    {
18        if (isalpha(text[i]))
19        {
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled1.exe
- Output Size: 129.787109375 KiB
- Compilation Time: 0.55s

Line: 40 Col: 2 Sel: 0 Lines: 40 Length: 807 Insert Done parsing in 0.172 seconds

Input/Output



```
C:\Users\josh\OneDrive\Desktop >
Enter plaintext: ATTACKATDAWN
Enter key: LEMON
Ciphertext: LXFOPVEFRNHR

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Process exited after 23.85 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the polyalphabetic substitution cipher was successfully implemented using a key.